# DIPU SINGH

Cloud & DevOps Engineer · Kubernetes · Infrastructure Automation

Guwahati, Assam, India | +91 6003239904 | dipudevops@outlook.com | GitHub | LinkedIn | Portfolio

## PROFESSIONAL SUMMARY

DevOps Engineer with ~4 years of experience building and maintaining production systems across AWS and Azure. Focused on improving deployment reliability, managing Kubernetes workloads, and automating infrastructure with Terraform. Experienced in handling real-world production challenges including scaling, deployment failures, and system monitoring. Comfortable taking full ownership of systems and working independently in fast-paced environments.

## TECHNICAL SKILLS

| | |
|---|---|
| **Cloud** | AWS (EC2, S3, Lambda, VPC, RDS, EKS, IAM, CloudWatch) · Azure (AKS, Functions, Blob Storage, VMs, Security Center) |
| **Containers** | Docker · Kubernetes (AKS, EKS) · Helm Charts · Azure Container Registry |
| **IaC** | Terraform · AWS CloudFormation · Ansible |
| **CI/CD** | GitLab CI/CD · Azure DevOps · GitHub Actions · ArgoCD |
| **Monitoring** | Prometheus · Grafana · CloudWatch · Azure Monitor |
| **Security** | DevSecOps · IAM · Vulnerability Remediation · Azure Security Center |
| **Scripting** | Bash · Python · Git |
| **MLOps (Basic)** | Docker-based ML deployments · containerized inference pipelines |

## PROFESSIONAL EXPERIENCE

**DevOps Engineer** · *Insightful IT Services LLP* *Nov 2023 – Present*

- Architected end-to-end CI/CD pipelines for .NET and Angular applications using GitLab CI/CD, cutting average deployment time by 90% and enabling zero-downtime blue-green releases across Dev, Staging, and Production environments.
- Automated Azure cloud infrastructure provisioning using Terraform, managing 50+ resources including virtual machines, storage accounts, virtual networks, NSGs, and security groups, eliminating manual provisioning errors entirely.
- Led migration of legacy monolithic applications to Kubernetes (AKS), implementing Horizontal Pod Autoscaler and right-sizing CPU/memory limits to reduce server costs by 40% and improve horizontal scalability under traffic spikes.
- Designed and deployed serverless event-driven workflows using Azure Functions integrated with Service Bus and Blob Storage triggers, cutting operational overhead by 60% and removing dependency on always-on VMs.
- Conducted security audits for 10+ applications using Azure Security Center and static analysis tooling, identifying and remediating 30+ vulnerabilities to achieve 100% CIS benchmark compliance.
- Automated database deployment pipelines using Azure DevOps release gates and SQL migration scripts, eliminating manual deployment errors and reducing release cycle time by 50%.
- Implemented centralised secrets management using Azure Key Vault across all environments, removing hardcoded credentials from 15+ pipelines and improving security posture significantly.

- Built and maintained Prometheus + Grafana monitoring dashboards with alerting rules, enabling early detection of resource saturation, pod crashloops, and pipeline failures before they impacted users.
- Designed and tested rollback strategies including Helm chart versioning and deployment checkpoints, reducing mean recovery time from failed releases from hours to under 10 minutes.
- Coordinated release windows with development and QA teams, managing environment-specific configuration promotion across three tiers and maintaining a consistent deployment runbook.

**DevOps Engineer (Contract)** · *Wingman Partners* *Jul 2023 – Aug 2023*

*Short-term contract engagement to support client infrastructure delivery and containerisation projects.*

- Deployed Dockerized ML-based theft-detection systems for retail clients, reducing infrastructure setup time by 50% through fully automated containerised deployment pipelines.
- Developed reusable, parameterised Terraform modules for AWS EC2, S3, and VPC provisioning, standardising Infrastructure as Code practices and cutting per-project setup time across 5+ client engagements.
- Managed on-premises Kubernetes clusters, tuning pod scheduling, resource quotas, and node affinity rules to reduce application latency by 25% under peak load.
- Debugged and resolved containerisation issues in client-deployed Docker environments, ensuring consistent runtime behaviour across dev and production parity configurations.
- Improved deployment scripts with idempotency checks and error handling, reducing failed deployments during client handover and enabling self-service re-runs by developers.

**DevOps Engineer** · *Trusnetix Technologies* *Nov 2020 – Feb 2022*

*Early-career role focused on Linux server operations, AWS infrastructure support, and CI/CD foundations in an on-prem and cloud hybrid environment.*

- Managed and maintained Linux-based production and staging servers, handling OS-level configuration, process supervision, user access, and scheduled maintenance tasks.
- Supported AWS infrastructure setup and day-to-day operations covering EC2 instance provisioning, S3 bucket management, VPC networking, and IAM role and policy administration.
- Built and maintained CI/CD pipelines for internal tools and applications, automating build, test, and deploy stages and reducing manual release effort for the engineering team.
- Monitored services using logging and alerting tools; performed log analysis, diagnosed production issues, and restarted failed services — reducing mean time to recovery on recurring incidents.
- Created runbooks and deployment checklists for routine operations, enabling consistent handoffs between shifts and reducing repeat incidents caused by undocumented steps.
- Assisted senior engineers with infrastructure upgrades and capacity planning exercises, gaining hands-on exposure to networking fundamentals, load balancing, and storage configuration.
- Participated in on-call rotations, handling basic server incidents independently and escalating complex issues with detailed root-cause context to senior engineers.

## KEY PROJECTS

### Multi-Cloud Kubernetes Cluster Automation

- Engineered Terraform templates to deploy highly available EKS and AKS clusters, enabling seamless cross-cloud workload scaling and disaster recovery.
- Integrated Prometheus + Grafana observability stack, reducing mean incident response time by 35%.

### Serverless CI/CD Pipeline for Azure Functions

- Architected an Azure DevOps pipeline automating testing and deployment of 11+ serverless functions, increasing release frequency by 70% while maintaining quality gates.

### Terraform-Driven AWS Infrastructure Automation

- Automated provisioning of 20+ AWS resources (VPC, EC2, RDS, S3, IAM) with modular Terraform configs, cutting manual setup time by 80%.

**ML Application Deployment (Basic MLOps)**

- Containerised a machine learning application using Docker and set up a repeatable deployment workflow for inference serving.

## EDUCATION

**Bachelor of Technology – Computer Science** *CGPA 7.44*
*Central Institute of Technology Kokrajhar*

**Diploma – Computer Science** *CGPA 7.2*
*Central Institute of Technology Kokrajhar*

## ACHIEVEMENTS & RECOGNITION

- Hackathon Winner – Startup Assam (Tezpur University): Built an AR-based teaching application recognised for innovation in educational technology.
- Bug Bounty Hunter: Acknowledged by Google, Seagate, and Dream11 for identifying and responsibly disclosing critical security vulnerabilities.

## LANGUAGES

**English** · **Hindi** · **Assamese**